ISOMORPHISMS DESCEND TO A FINITE LEVEL IN THE DESCENT SITUATION

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Flat closed immersions of finite presentation. Morphisms, Lemma morphisms-lemma-flat-closed-immersions-finite-presentation
- Limit contained in constructible. Limits, Lemma lemma-limit-contained-in-constructible (same chapter)

Source reference. Original reference 081D: sources/stacks-chow/tags.txt, source lines 10213–10213. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 07RP: sources/stacks-chow/tags.txt, source lines 9874–9874. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZP: sources/stacks-chow/tags.txt, source lines 2602–2602. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0819: sources/stacks-chow/tags.txt, source lines 10209–10209. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 05F4: sources/stacks-chow/tags.txt, source lines 7013–7013. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Notation and assumptions as in Situation 081D. If*

(1) *f is an isomorphism, and*
(2) *f_0 is locally of finite presentation,*

then f_i is an isomorphism for some $i \geq 0$.

Proof. By Lemmas 07RP and 01ZP we can find an i such that f_i is flat and a closed immersion. Then f_i identifies X_i with an open and closed subscheme of Y_i, see Morphisms, Lemma 0819. By assumption the image of $Y \to Y_i$ maps into $f_i(X_i)$. Thus by Lemma 05F4 we find that $Y_{i'}$ maps into $f_i(X_i)$ for some $i' \geq i$. It follows that $X_{i'} \to Y_{i'}$ is surjective and we win. □

Date: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.